Static and kinetic friction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish adjustable static friction from sliding friction.

SUCCESS CRITERIA - I can...

1. Compare a push with the static-friction limit.
2. Find actual friction for rest or sliding.
3. Explain friction direction using relative slipping.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A spacecraft of mass **3 kg** pushes an asteroid of mass **300 kg** with **15 N**. **Compare interaction forces** and the accelerations due to that interaction.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: 0 ≤ f_s ≤ μ_s N; f_k = μ_k N.
s = static; k = kinetic; N = normal force. Static friction adjusts up to its maximum. Kinetic friction applies during sliding.

Worked reasoning: A box remains still while pushed with **3 N**. Static friction is **3 N** opposite the push, provided its maximum is at least **3 N**. Static friction need not equal its maximum.

Your turn: Draw the horizontal forces. Then redraw for a **5 N** push that is still below the maximum.

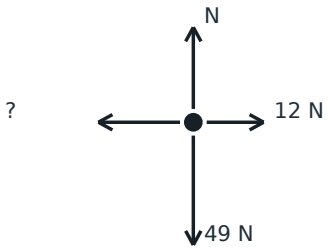


Box remains at rest. Redraw for a 5 N push.

Marzano check - circle: ① ② ③

03 Actual versus maximum

A **5 kg** box is initially at rest on a level floor. Its friction coefficients are μ_s = **0.40** and μ_k = **0.30**. A horizontal push is **12 N**.
1. Use g = **9.8 N/kg**.
2. **Find the normal force**, maximum static friction, actual friction and acceleration.
3. **Label the diagram**.



The dot represents the object. Initially at rest

Marzano check - circle: ① ② ③

04 Sliding case

Increase the push on that box to **25 N**. **Find friction** and acceleration after it slides.

Marzano check - circle: ① ② ③

Static and kinetic friction

antocciphysics

Continue down each column.

05 Estimate from data

A class tests a block with a **20 N** normal force.
1. Use the repeated measurements to **find mean peak force**, mean sliding force, and estimates of μ_s and μ_k.
2. Which estimate describes the threshold for slipping?

Trial	Peak force (N)	Sliding force (N)
1	8	6
2	9	5
3	7	7

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

x_i=**-4±0.5 m**, v=**6±0.1 m/s** and exact t=**8 s**. **Give central final position**, its conservative interval and full width. **Name something you** are treating as true. **Label the diagram** with the given positions and any prediction.

Position (m)



Time (s)

Marzano check - circle: ① ② ③

07 Exit check

A **10 kg** box has μ_s = **0.50** and μ_k = **0.40**.
1. Use g = **9.8 N/kg**.
2. **Compare a 30 N** push with a **60 N** push, starting from rest each time.
3. **Find actual friction** and acceleration in both cases.

Push (N)	Still or sliding?	Friction (N)	a (m/s²)
30	___	___	___
60	___	___	___

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **5 kg** box has μ_s=**0.4** and μ_k=**0.3** on a level floor; use g=**10 N/kg**. A horizontal **22 N** pull begins from rest. **Find maximum static friction**, acceleration after motion starts, and the pull needed for steady sliding.

Add and label force arrows.

Draw the sliding case, with actual friction.

Marzano check - circle: ① ② ③